ELECTRIFICATION AND NET-ZERO TRANSITION IN NEPAL: BRIDGING THE GAP BETWEEN DEMAND AND GENERATION

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ABSTRACT

Renewable energy accounts for about 29.1% of the world's total electricity production. Of that, 15.3% is from hydropower. Nepal being rich in water resources with the estimated capacity of 42 GW energy has 98% of total electricity from hydropower, which stands as the country's dominant source of electricity generation. While the majority of countries still struggle for net zero emission targets and Nationally Determined Contribution (NDC), Nepal is uniquely positioned in the global energy transition with its predominantly clean electricity generation, largely based on hydropower. However, the energy demand tends to increase as per the increase in per capita, population growth, urbanization rate, industrialization and other sources of productive use of energy. In the context of Nepal, shifting towards electricity can also be considered as shifting towards net zero emission. Several government efforts like reducing tax on EV import, subsidies, investment in renewable projects or special loans for renewable energy, NDC target to 15,000 MW of electricity generation by 2030 and other energy policies are currently ongoing. Cooking shifting towards induction from LPG, kerosene and biomass, and transportation shifting towards electric vehicles are notable trends toward renewable energy and net zero. However, this electrification surge will significantly escalate electricity demand. This study quantifies the projected electricity demand based on NDC-aligned transition scenarios and compares it against current and planned generation capacities. The analysis highlights a critical gap between demand from the transport and cooking sectors that may outpace electricity supply if implementation of generation projects lags. To ensure a sustainable and resilient energy transition, coordinated planning, policy enforcement, and capacity expansion must be accelerated.

KEYWORDS: Electric Vehicles, Energy Transition, Electrification, Net-Zero, Nepal

INTRODUCTION

Nepal is at an unusual starting point in its energy transition. With 98% of its electricity already produced from hydropower and renewable sources (NEA, 2023/24), Nepal has the opportunity to become a leader in regional clean energy use. While many countries face the challenges of decarbonizing fossil-fuel heavy grids, Nepal has access to enormous hydropower; however,

the usage of imported fossil fuels for transportation and cooking remains significant. The national focus on petroleum product and LPG imports has contributed greatly to the trade deficit. Therefore, electrifying these sectors is not just an environmental issue; it is also a means of enhancing energy security and the broader economy.

Though Nepal's electricity generation is mostly renewable (nearly 100% hydropower), the overall energy mix shows a different picture. The share of electricity in Nepal's total energy mix was only about 5% in 2020, 13 times smaller than traditional biomass and one-sixth of fossil fuels. This indicates an important gap: achieving net-zero will take more than a clean grid; it requires shifting most energy consumption, especially cooking, heating, and transport, onto that grid.

The government of Nepal through its Nationally Determined Contributions (NDC) and applicable policies related to the energy sector, seeks to generate 15,000 MW of electricity by 2030 (Gov. of Nepal, 2025) which is well above the current generation capacity of 3,591 MW (NEA, 2025). This objective responds to the increasing domestic electricity demand as a result of economic development and changing consumption patterns. Technological interventions such as induction cooking and electric mobility, particularly electric two and three-wheelers, are critical milestones for Nepal's transition to net-zero emissions (The New York Times, n.d.). Recent adoption of new electric vehicles (EVs) results in more electricity consumption and demand, primarily because distribution, but not generation, is constrained. While the increased consumption from e-cooking is not shiftable from peak hours, the electricity demand created by EV charging consumption patterns has sufficient flexibility to enable charging in the off-peak period overnight. Charging overnight, at night when the 2-W and 3-W are charging, will improve and balance the load profile of hydropower and take advantage of excess run-of-river hydropower capacity at night (especially during the low or non-peak hours of the current daily load profile). NEA's own network of 62 public charging stations already averaged 35,000 units of electricity across 2,500 customers daily in FY2024/25, with individual station capacity utilisation of only 10–40%, indicating that EV charging is already concentrated in peak windows rather than distributed uniformly across the day (NEA, 2025).

The momentum behind electric mobility is stronger than generally recognized. A trolley bus was first operated in Nepal in 1975 with the aid of the Chinese government (The Annapurna Express, 2025). However, new data shows that electric vehicles comprised a remarkable 73% of the four-wheeled passenger cars imported in the fiscal year ending mid-2025. This is an early indication that market forces and consumer uptake will quickly progress, making accurate demand forecasting for the electricity sector critical to averting a supply crisis.

Electrification in the cooking sector is quite challenging. Nepal has 98% electricity access; however, the question arises whether the 98% electricity access is reliable (NEA 2023/24, n.d.). 98% of the population has electricity access for lighting purposes. On increasing the electricity demand by transitioning to electric cookstoves, the electricity demand significantly increases. This poses a major challenge for demand-side management, as the national load curve is dominated by residential load, which adds further load during peak hours and raises the transmission capacity required. NEA recorded a system peak demand of 2,901 MW against

a national peak of 2,409 MW in FY2024/25, with an annual load factor of only 66.89%, confirming that the pre-electrification grid already carries a significant peak-to-average spread (NEA, 2025).

Nepal's energy industry is mainly dependent on hydropower and hydropower greatly reduced output during the dry winter with widespread implications (Aryal et al., 2024; Singh et al., 2022). The NEA FY2024/25 energy balance confirms the scale of this asymmetry: domestic generation across the six dry-season months (Mangsir–Baishakh, roughly December–May) totalled approximately 4,692 GWh, compared with 9,622 GWh across the six wet-season months, and 1,571 GWh of the year's total 1,712 GWh in electricity imports were required to bridge the dry-season deficit (NEA, 2025). As EV and electric cooking grow, this seasonal misalignment could worsen supply constraints without sufficient investment in energy storage, diversified generation and trade with neighbouring countries, particularly India. Nepal must scale its generation and distribution capacities fast enough to keep up with the increasing demand required by electrifying sectors like cooking and transportation.

Extensive research has been conducted on both short-term and long-term electricity demand forecasting. For example, Shariff (2022) compared the performance of ARIMA and LSTM models for electricity demand prediction. Traditionally, several regression methods have been used for forecasting time series data (Jifri et al., 2017); however, deep learning models offer better accuracy by capturing nonlinear trends (Adebayo Olusegun Aderibigbe et al., 2023). The impact of technological interventions has not yet had a significant effect on electricity consumption due to being in a beginning phase of market penetration. Consequently, using LSTM models to predict energy demand from electric vehicles (EVs) and electric cooking can be challenging, as the neural network may not fully capture the intensity of these technological effects. To incorporate the impact of such interventions, a Sigmoid curve can be applied and its estimated demand added to the baseline demand forecasted using LSTM, resulting in a more accurate overall future electricity demand prediction (Asthana, 1995). This paper aims to evaluate the likely demand scenarios through 2030 and 2045 by combining LSTM-based forecasting with penetration estimates of electric cooking and EV technologies.

MATERIALS AND METHODS

The initial phase of the study began with the collection of Gross Domestic Product (GDP), Total Population, Urban Population, per capita income, per capita electricity consumption, and total electricity consumption (World Bank, 2025). The data was analysed to predict electricity consumption forecast in Nepal until 2030. A Multivariate Long Short-Term Memory (LSTM) was used to conduct the prediction (Ghanbari & Borna, 2021; Waqas & Humphries, 2024). The model data was trained in 2001-2021, and subsequently it was used to predict the data from 2021-2030.

Simultaneously an energy supply forecast is needed for completed energy demand forecast including the unconditional target in conformity with the Nationally Determined Contribution 3.0. The previous methodology assumes several technology interventions, such as induction cookstoves and the introduction of electric vehicles into the Nepali market. However,

electricity demand will be forecasted using the earlier LSTM forecast as a baseline and then adding the estimated 30%, 40%, and 50% market penetration of these technologies. The entire methodology will be utilized to predict energy demand up to 2030.

RESULTS

The forecast from the LSTM model predicted the total energy demand will reach 14,690 GWh by 2030 and 16,120 GWh by 2045 considering business-as-usual condition. The business-as-usual model seems to saturate after 2035 with a very low growth rate.

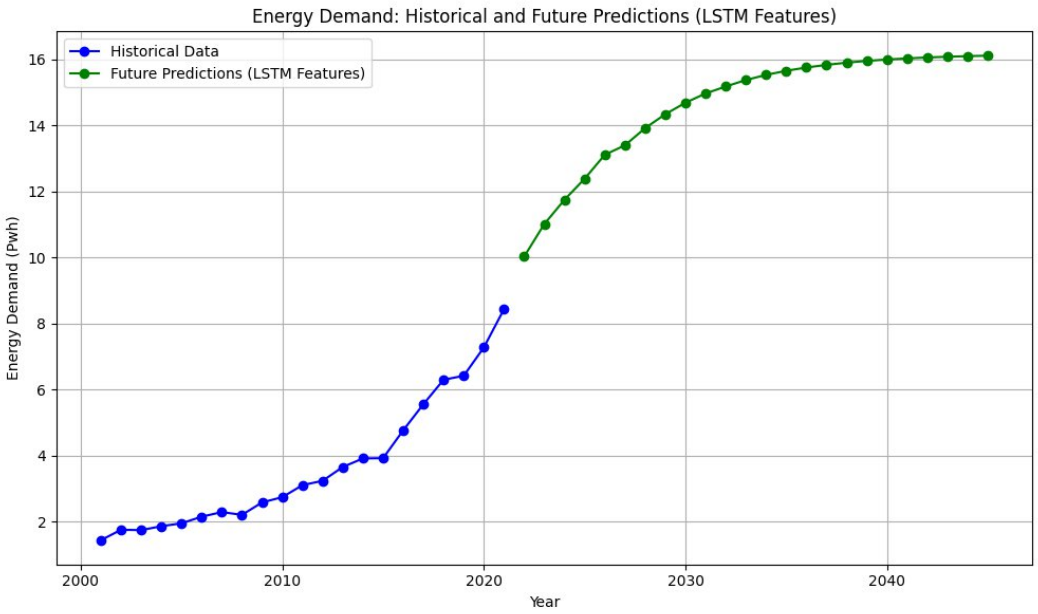


Figure 1. Baseline Prediction of Energy Demand for Business-as-Usual Condition

The figure below represents the unconditional target set by the Government of Nepal as Nationally Determined Contribution 3.0 to generate 35,000 GWh of electricity at 40% capacity factor by 2035. It creates the foundation of electricity supply in this research. Linear trend is used to connect the data of electricity generation. However, the unconditional target seems to exceed the electricity demand till 2030. Several hydropower projects have received construction licenses to meet the NDC 3.0 target (DOED, 2025).

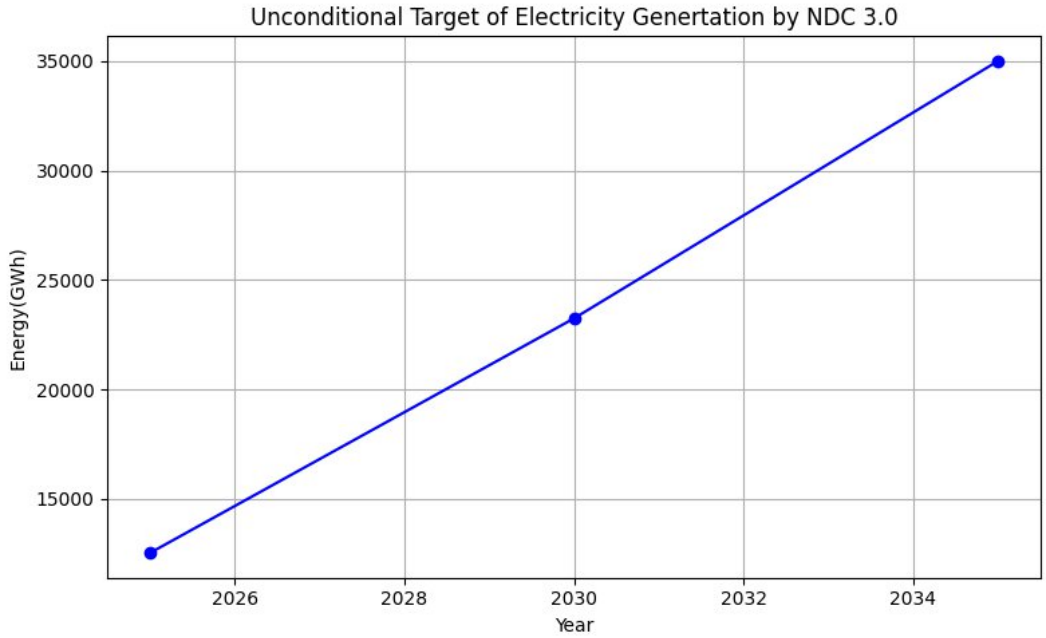


Figure 2. Unconditional Target Set by NDC 3.0

The fossil fuel pattern seems to be growing slowly for the past few years compared to earlier. This indicates the effect of EV penetration in the transportation sector. The sudden drop in fuel consumption in 2020 shall be considered as the effect of the lockdown. However, the demand further seems to increase but not at a higher rate compared to earlier data. The figure below has been forecasted based on the trend with the LSTM model, which also reflects a very slow growth rate.

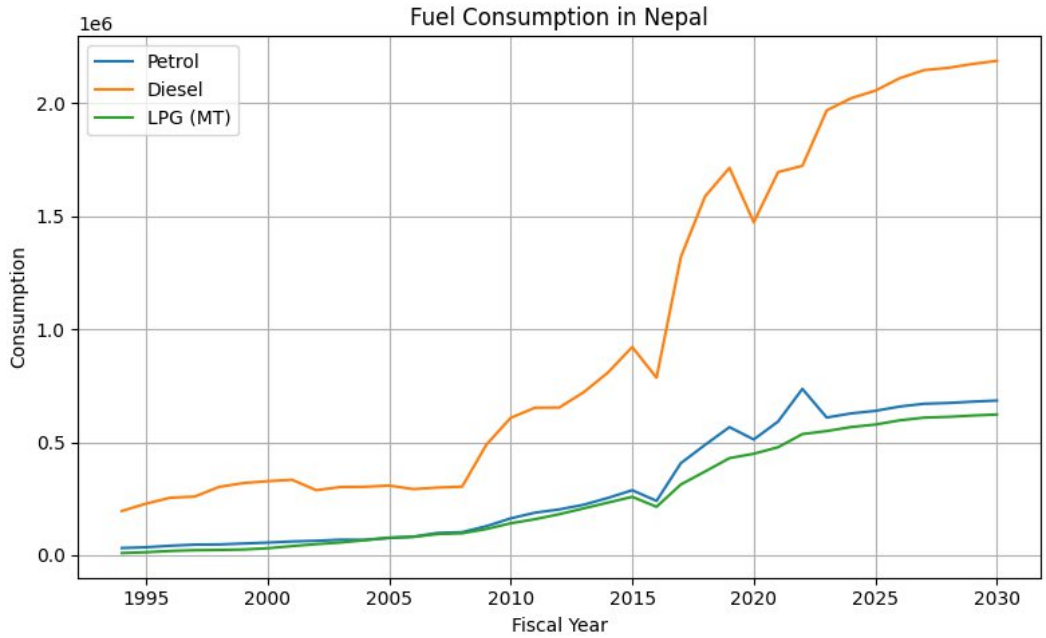


Figure 3. Fossil fuel consumption pattern since 1995 to 2022 (NOC, 2025)

The potential demand increases in electricity due to transition of fossil fuels (petrol + diesel) to electricity remains 2,660 GWh assuming current energy consumption scenario in transportation sector. The 7.63 GWh is the current electricity demand due to EV penetration.

However, on calculating equivalent electric energy for total transportation energy use, the demand escalates to 2,660 GWh. Assuming 100% electrification by 2045, a sigmoid curve (Asthana, 1995) is traced to mimic the behaviour of technology intervention to 2,660 GWh for 2045, from 7.63 GWh for 2024. The smooth sigmoid curve data is then used to add up the demand in the final prediction curve.

Similarly, electric cooking penetration curve was forecasted based on a sigmoid curve for 76.46 GWh in 2024 and 14,417.33 GWh in 2050 assuming 100% electrification by 2050. The energy demand for cooking is considerably higher and transition to electric cooking may be much slower because of ease of use of the traditional cooking methods and electric cooking has higher initial cost and still 51% of population use fuel wood and 44.3 % of population use LPG (Household | National Population and Housing Census 2021 Results, n.d.) which creates equal opportunity and challenges for the transition to electric cooking.

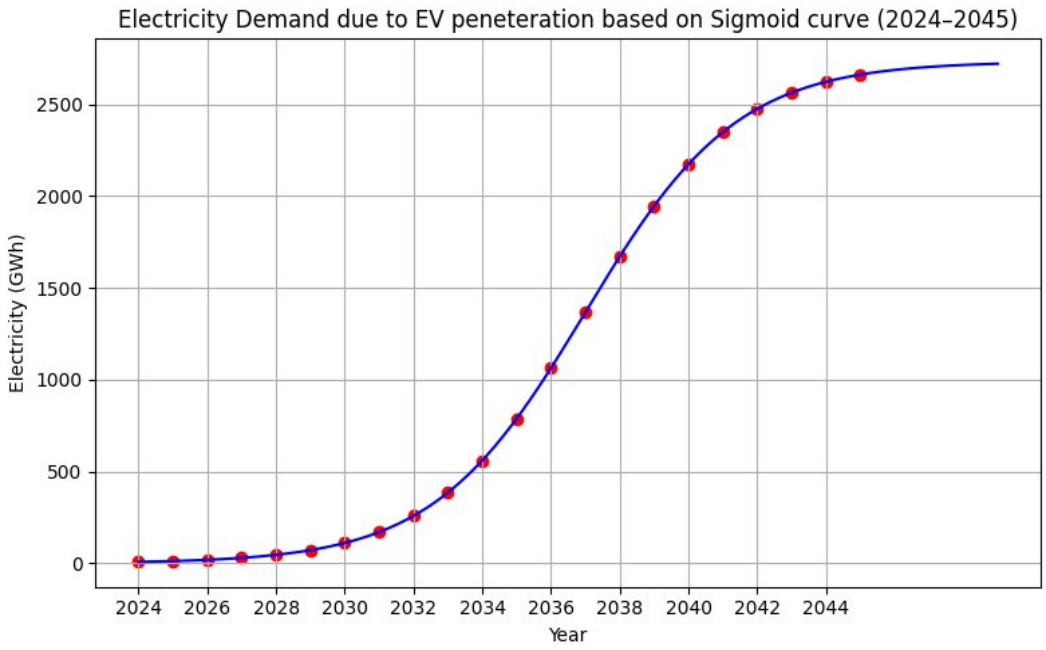


Figure 4. Electricity Demand Due to EV Penetration

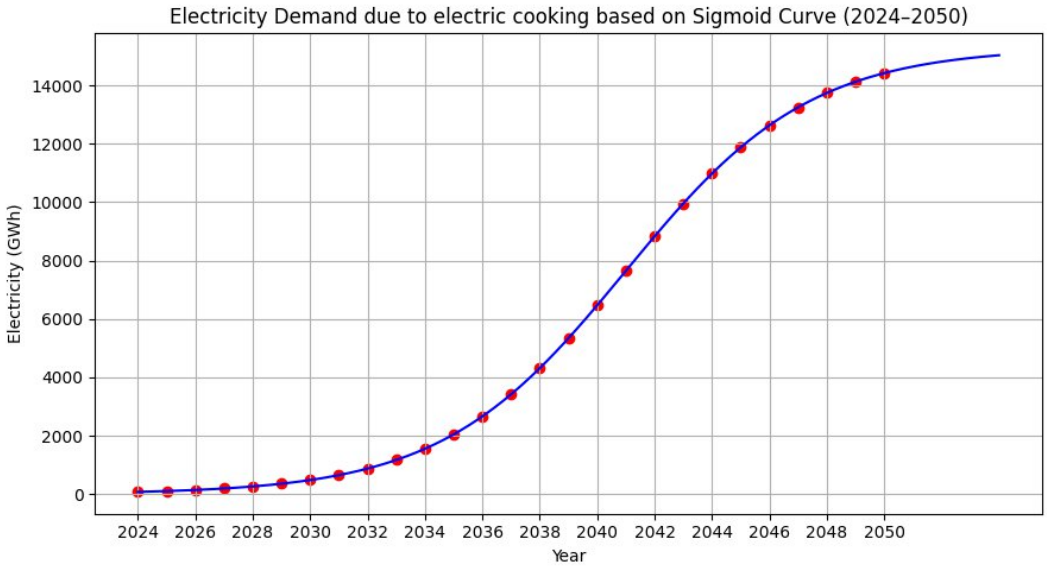


Figure 5. Electricity Demand Due to Electrification of Cooking

The forecast analyses show that the Multivariate Long Short-Term Memory (LSTM) model trained on historical socioeconomic and energy consumption from 2001-2021 estimates the total electricity demand in Nepal to be 16,270 GWh in 2030. This total demand was generated using the data shown in Figure 5 incorporating the trend in GDP, population, urbanization, per capita income, and electricity consumption trends.

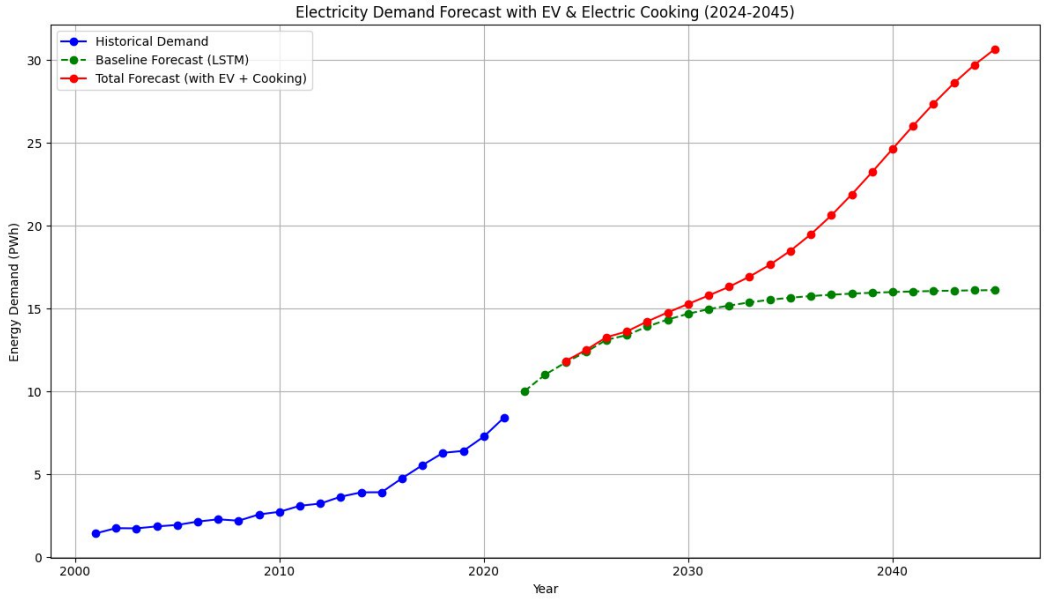


Figure 6. Total Expected Electricity Demand (Baseline + EV + Cooking)

In addition to this baseline demand, the analysis offers estimates of additional electricity demand anticipated to result from the market penetration of electric vehicles (EVs) and induction cookstoves. Figure 4 shows the negative correlation between fossil fuel consumption and electricity demand associated with EV adoption. This suggests a radical transformation has the potential to occur in the energy sector as EV consumption increases. Additionally, Figures 5 and 6 display the anticipated increasing electricity consumption from these sectors from the sigmoid penetration curves, representing realistic adoption rates over time.

Incorporating these technological changes into the electricity demand forecast, shows a substantially higher increase in electricity demand compared to the LSTM baseline. This complete forecast assumed percentages for market penetration of EVs and electric cooking. For context, the total electricity demand is estimated to reach approximately 16,270 GWh by 2030, and will continue to grow substantially towards 2045.

On the supply side, the respective energy supply forecast, based on Nepal's Nationally Determined Contribution (NDC) 3.0 unconditional target, is for installed capacity of electricity generation of 15,000 MW by 2030 Figure 2. 15,000 MW represents roughly, under a theoretical 40% capacity factor, 35,000 GWh annually.

DISCUSSION AND CONCLUSIONS:

The LSTM baseline forecast indicates that electricity demand in Nepal will reach around 14,690 GWh by 2030, in a business-as-usual scenario, with reduced demand growth after 2035. In comparison, the unconditional NDC 3.0 target of 35,000 GWh of electricity by 2035 implies that supply would have initially exceeded demand. When electricity demand from electric vehicle penetration and electric cooking using sigmoid adoption curves is added, however, total demand increases rapidly, reaching approximately 16,270 GWh by 2030 and continuing to increase rapidly towards 2045. This shows the strong influence of technology-led electrification on future demand, and the associated need for proactive planning of supply increases. Even though the forecasted supply increase trajectory is hydropower-based and a good start for a clean energy transition, if the electrification targets are met with higher penetration rates the hydropower supply could lag behind the increased demand. A large share of hydropower could also increase the likelihood of seasonal shortages, particularly during the winter months.

While the demand projections presented here provide a macro-level picture of electrification trajectories, several grid-level dimensions require further analysis to fully validate the conclusions. First, peak demand assessment is critical: induction cooking loads are largely non-shiftable and coincide with existing morning and evening residential peak hours, which may require significant transmission capacity upgrades beyond what the annual energy figures suggest; NEA's recorded system peak of 2,901 MW against a national peak of 2,409 MW in FY2024/25, with a load factor of 66.89%, illustrates the existing peak-to-average spread even before large-scale cooking electrification loads are added. Second, Nepal's hydropower system is predominantly run-of-river (ROR), with dry-season output typically falling to 30-40% of monsoon-season capacity, confirmed by the NEA FY2024/25 energy balance, which recorded only 4,692 GWh of domestic generation across the six dry-season months versus 9,622 GWh in the six wet-season months, requiring 1,571 GWh of imports to cover the gap; reserve margin analysis accounting for this seasonal variability is needed to confirm supply adequacy under the 30-50% electrification penetration scenarios modelled. Third, EV charging coincidence factors; the probability that a large fraction of the two and three-wheeler fleet charges simultaneously, must be quantified to assess distribution-level loading; while overnight off-peak charging is feasible in principle, grid infrastructure in Nepal's distribution network may limit practical load shifting; NEA's 62 public charging stations averaged 35,000 units of electricity daily in FY2024/25 at only 10-40% capacity utilisation, suggesting existing charging demand is already concentrated in peak windows rather than spread across off-peak hours. These grid balancing and reliability dimensions are identified as important areas for future work, and the conclusions of this paper should be read in that context.

In conclusion, despite Nepal having an excellent opportunity to move toward a net-zero energy system, whether Nepal can meet these targets will depend on timely and integrated planning where expansions of electricity supply and electricity demand are coordinated, due to the accelerated demand from increasing adoption of EV and electric cooking.

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